

Sample-Optimal Parametric Q-Learning Using Linearly Additive Features

-- ICML2019

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2021.04.07

Tabular case without function approximation

- Value iteration
- Q -learning

Tabular case with function approximation

- Phased Parametric Q -Learning
- Optimal Phased Parametric Q -Learning

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Value Iteration

Value Iteration, for estimating $\pi \approx \pi_*$

Algorithm parameter: a small threshold $\theta > 0$ determining accuracy of estimation

Initialize $V(s)$, for all $s \in \mathcal{S}^+$, arbitrarily except that $V(\text{terminal}) = 0$

Loop:

| $\Delta \leftarrow 0$

| Loop for each $s \in \mathcal{S}$:

| $v \leftarrow V(s)$

| $V(s) \leftarrow \max_a \sum_{s',r} p(s', r | s, a) [r + \gamma V(s')]$

| $\Delta \leftarrow \max(\Delta, |v - V(s)|)$

until $\Delta < \theta$

Output a deterministic policy, $\pi \approx \pi_*$, such that

$$\pi(s) = \operatorname{argmax}_a \sum_{s',r} p(s', r | s, a) [r + \gamma V(s')]$$

Q-Learning

Q-learning (off-policy TD control) for estimating $\pi \approx \pi_*$

Algorithm parameters: step size $\alpha \in (0, 1]$, small $\varepsilon > 0$

Initialize $Q(s, a)$, for all $s \in \mathcal{S}^+, a \in \mathcal{A}(s)$, arbitrarily except that $Q(\text{terminal}, \cdot) = 0$

Loop for each episode:

Initialize S

Loop for each step of episode:

Choose A from S using policy derived from Q (e.g., ε -greedy)

Take action A , observe R, S'

$Q(S, A) \leftarrow Q(S, A) + \alpha [R + \gamma \max_a Q(S', a) - Q(S, A)]$

$S \leftarrow S'$

until S is terminal

Sample Complexity



In the tabular setting without any feature, the sample complexity lower bound is known to be $\Omega(\frac{SA}{(1-\gamma)^3\epsilon^2})$.

What if states and actions get much bigger? Function approximation!

Tabular case without function approximation

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Tabular case with function approximation

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- Optimal Phased Parametric Q -Learning

Function Approximation



Feature Vector

Generative Model

Representative States

Feature Vector



Suppose that the learning agent is given a set of K feature functions $\phi_1, \phi_2, \dots, \phi_K: \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$. The feature ϕ maps the raw state and action (s, a) into the K -dimensional vector

$$\phi(s, a) = [\phi_1(s, a), \phi_2(s, a), \dots, \phi_K(s, a)] \in \mathbb{R}^K$$

Property of Feature Vector



Expressing Ability

Suppose the feature vector $\phi(s, a)$ is sufficient to express the future dynamics of the process conditioning on the current raw state and action.

Definition 1 (Feature-based Linear Transition Model). Consider a DMDP instance $M = (\mathcal{S}, \mathcal{A}, P, r, \gamma)$ and a feature map $\phi: \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^K$. We say that M admits a linear feature representation ϕ if for every s, a, s' ,

$$P(s'|s, a) = \sum_{k \in [K]} \phi_k(s, a) \psi_k(s').$$

for some functions $\psi_1, \dots, \psi_K: \mathcal{S} \rightarrow \mathbb{R}$. We denote the set of all such MDP instances as $\mathcal{M}^{trans}(\mathcal{S}, \mathcal{A}, \gamma, \phi)$, and $\mathcal{M}_K^{trans}(\mathcal{S}, \mathcal{A}, \gamma)$ the set of all DMDP instances that admits a K -dimensional feature representation.

Property of Feature Vector



Why not express Q -function or value directly?

If the MDP admits a linear transition model using ϕ , the Q -functions admit a linear model.

Proposition 1. Let $M \in \mathcal{M}^{trans}(\mathcal{S}, \mathcal{A}, \gamma, \phi)$, Then $Q^\pi \in \text{Span}(r, \phi)$ for all π .

Property of Feature Vector



Why not express Q -function or value directly?

If the MDP admits a linear transition model using ϕ , the Q -functions admit a linear model.

Given the feature map $\phi: \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^K$, we can parameterize the Q -functions, value functions and policies using $\omega \in \mathbb{R}^K$ by

$$\begin{aligned} Q_\omega(s, a) &:= r(s, a) + \gamma \phi(s, a)^T \omega \\ V_\omega(s) &:= \max_{a \in \mathcal{A}} Q_\omega(s, a) \\ \pi_\omega(s) &:= \operatorname{argmax}_{a \in \mathcal{A}} Q_\omega(s, a) \end{aligned}$$

Generative Model



Where samples come from? Generative Model.

Suppose that we are given a generative model where the agent is able to query transition samples and reward from any state-action pair $(s, a) \in \mathcal{S} \times \mathcal{A}$.

We aim to learn a near-optimal policy using a small number of samples, which depends on K but not S .

Representative States



How to sample? The state-action pairs cannot be too alike.

Assumption 1 (Representative States and Regularity of Features). There exists a representative state-action set $\mathcal{K} \subset \mathcal{S} \times \mathcal{A}$ with $|\mathcal{K}| = K$ and a scalar $L > 0$ such that

$$\|\phi(s, a)^T \Phi_{\mathcal{K}}^{-1}\|_1 \leq L, \forall (s, a)$$

where $\Phi_{\mathcal{K}} \in \mathbb{R}^{K \times K}$ is the collection of row feature vectors $\phi(s, a)$ where $(s, a) \in \mathcal{K}$ and $L \geq 1$.

Sample Complexity Lower Bound



Theorem 1 (Sample Complexity Lower Bound). *Let $M = (\mathcal{S}, \mathcal{A}, P, r, \gamma)$ be an instance of DMDP, and let $\mathcal{A}$ be any algorithm that queries sample transitions of M and outputs a policy. Let $\pi^{\mathcal{A}, M, N}$ be the output of $\mathcal{A}$ using N samples. Then*

$$\inf_{\mathcal{A}} \sup_{M \in \mathcal{M}_K^{trans}(\mathcal{S}, \mathcal{A}, \gamma)} \mathbb{P} \left(\sup_{s \in \mathcal{S}} (v^*(s) - v^{\pi^{\mathcal{A}, M, N}}(s)) \geq \epsilon \right) \\ \geq 1/3, \quad \text{if } N = O \left(\frac{K}{(1 - \gamma)^3 \cdot \epsilon^2 \cdot \log \epsilon^{-1}} \right),$$

provided $\epsilon \leq \epsilon_0$ for some $\epsilon_0 \geq 0$.

Theorem 1 suggests that, in order to solve the feature-based MDP to precision level ϵ with probability at least $2/3$, any algorithm needs at least $\tilde{\Omega}(\frac{K}{(1-\gamma)^3 \epsilon^2})$ sample transitions in the worst case.

Phased Parametric Q-Learning

Algorithm 1 Phased Parametric Q-Learning (PPQ-Learning)

- 1: **Input:** A DMDP $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, r, \gamma)$ with a generative model
 - 2: **Input:** Integer $N > 0$
 - 3:
 - 4: **Initialize:** $R \leftarrow \Theta \left[\frac{\log N}{1-\gamma} \right]$, $w \leftarrow 0 \in \mathbb{R}^K$;
 - 5: **Repeat:**
 - 6: **for** $t = 1, 2, \dots, R$ **do**
 - 7: Pick a representative set $\mathcal{K} \subset \mathcal{S} \times \mathcal{A}$ satisfying (4).
 - 8: $Q \leftarrow 0 \in \mathbb{R}^K$;
 - 9: **for** $(s, a) \in \mathcal{K}$ **do**
 - 10: Obtain $\frac{N}{KR}$ samples $\{s^{(j)}\}$ i.i.d. from $P(\cdot|s, a)$;
 - 11: $Q[(s, a)] \leftarrow \frac{KR}{N} \sum_{j=1}^{N/KR} \Pi_{[0, (1-\gamma)^{-1}]} [V_w(s^{(j)})]$;
 - 12: $\triangleright \Pi_{[a, b]}$ projects a number onto $[a, b]$
 - 13: **end for**
 - 14: $w \leftarrow \Phi_{\mathcal{K}}^{-1} Q$;
 - 15: **end for**
 - 16: **Output:** $w \in \mathbb{R}^K$
-

Error Bound and Sample Complexity



Theorem 2 (Convergence of Algorithm 1). *Suppose Assumption 1 holds. Suppose that the DMDP instance $M = (\mathcal{S}, \mathcal{A}, P, r, \gamma)$ has an approximate transition model $\tilde{P}$ that admits a linear feature representation ϕ (Defn. 1), such that for some $\xi \in [0, 1]$, $\|P(\cdot | s, a) - \tilde{P}(\cdot | s, a)\|_{TV} \leq \xi$, $\forall (s, a)$. Let Algorithm 1 takes $N > 0$ samples and outputs a parameter $w \in \mathbb{R}^K$. Then, with probability at least $1 - \delta$, $\|v^{\pi_w} - v^*\|_\infty$*

$$\leq L \cdot \left(\sqrt{\frac{K}{N \cdot (1 - \gamma)}} + \xi \right) \cdot \frac{\text{poly log}(NK\delta^{-1})}{(1 - \gamma)^3}.$$

(Sample complexity). When the MDP is fully realizable under the features, $\xi = 0$. Then the number of samples needed for achieving ϵ policy error is

$$\tilde{\Omega}\left[\frac{KL^2}{(1 - \gamma)^7 \epsilon^2}\right]$$

Error Bound and Sample Complexity



(Sample complexity). When the MDP is fully realizable under the features, $\xi = 0$. Then the number of samples needed for achieving ϵ policy error is

$$\tilde{\Omega}\left[\frac{KL^2}{(1-\gamma)^7\epsilon^2}\right]$$

Still far from the sample complexity lower bound.

Theorem 1 suggests that, in order to solve the feature-based MDP to precision level ϵ with probability at least $2/3$, any algorithm needs at least $\tilde{\Omega}\left(\frac{K}{(1-\gamma)^3\epsilon^2}\right)$ sample transitions in the worst case.

More Techniques to Achieve the Lower Bound



Monotonicity preservation

Variance reduction

Monotonicity Preservation



Monotonicity
property:

$$V_\theta \leq \mathcal{T}_{\pi_\theta} V_\theta \leq \mathcal{T}_{\pi_\theta}^2 V_\theta \leq \dots \leq \mathcal{T}_{\pi_\theta}^\infty V_\theta \leq v^{\pi_\theta} \leq v^*, \text{ pointwise.}$$

Assumption 2 (Anchor State-Action Pairs). *There exists a set of anchor state-action pairs $\mathcal{K}$ such that for any $(s, a) \in \mathcal{S} \times \mathcal{A}$, its feature vector can be represented as a convex combination of the anchors $\{(s_k, a_k) \mid k \in \mathcal{K}\}$:*

$$\exists \{\lambda_k\} : \phi(s, a) = \sum_{k \in \mathcal{K}} \lambda_k \phi(s_k, a_k), \quad \sum_{k \in \mathcal{K}} \lambda_k = 1, \lambda_k \geq 0.$$

Monotonicity Preservation



Parametrization. Uses a collection of parameters $\theta = \{\omega^{(i)}\}_{i=1}^Z$ instead of a single vector, with $Z = \tilde{O}\left(\frac{1}{1-\gamma}\right)$. And parameterized policy and value functions take the form

$$V_{\theta}(s) := \max_{h \in [Z]} \max_{a \in \mathcal{A}} (r(s, a) + \gamma \phi(s, a)^T \cdot \omega^{(h)})$$
$$\pi_{\theta}(s) \in \operatorname{argmax}_{a \in \mathcal{A}} \max_{h \in [Z]} (r(s, a) + \gamma \phi(s, a)^T \cdot \omega^{(h)})$$

Confidence bound $\epsilon^{(i,j)}$.

$$\bar{\omega}^{(i,0)}(k) \leftarrow \max\{0, \min\{\omega^{(i,0)}(k) - \epsilon^{(i,0)}(k), (1 - \gamma)^{-1}\}\}$$

Then we have

$$\phi(s, a)^T \bar{\omega}^{(i,j)} \leq P(\cdot | s, a)^T V^{(i,j-1)} \leq P(\cdot | s, a)^T V^{(i,j)}$$

Variance Reduction



To obtain an entry $Q^{(i,j)}(s, a)$ of the new Q -function, we need to estimate $P(\cdot | s, a)^T V_{\theta^{(i,j-1)}}$ with sufficient accuracy, so we split it into two parts.

$$P(\cdot | s, a)^T V_{\theta^{(i,j-1)}} = P(\cdot | s, a)^T V_{\theta^{(i,0)}} + P(\cdot | s, a)^T (V_{\theta^{(i,j-1)}} - V_{\theta^{(i,0)}})$$

Outer loop to estimate $P(\cdot | s, a)^T V_{\theta^{(i,0)}}$

Inner loop to estimate the offset $P(\cdot | s, a)^T (V_{\theta^{(i,j-1)}} - V_{\theta^{(i,0)}})$

Optimal Phased Parametric Q-Learning



Algorithm 2 Optimal Phased Parametric Q-Learning (OPPQ-Learning)

```

1: Input: A DMDP  $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, r, \gamma)$  with anchor state-action pairs  $\mathcal{K}$ ; feature map  $\phi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ ;
2: Input:  $\epsilon, \delta \in (0, 1)$ 
3: Output:  $\theta \in \mathbb{R}^K$  with  $|\theta| = \Theta[(1 - \gamma)^{-1} \log^2 \epsilon^{-1}]$ 
4:
5: Initialize:  $R' \leftarrow \Theta(\log[\epsilon^{-1}(1 - \gamma)^{-1}])$ ,  $R \leftarrow \Theta[R'(1 - \gamma)^{-1}]$   $\triangleright$  initialize the numbers of iterations
6:  $\{w^{(i,j)}, \epsilon^{(i,j)}, \bar{w}^{(i,j)}\}_{i \in [0, R'], j \in [0, R]} \subset \mathbb{R}^K$  as 0 vectors  $\triangleright$  initialize parameters
7:  $m \leftarrow C \cdot \frac{1}{\epsilon^2} \cdot \frac{\log(R'RK\delta^{-1})^{4/3}}{(1-\gamma)^3}$ ,  $\triangleright$  mini-batch size for outer loop
    $m_1 \leftarrow C \cdot \frac{\log(R'RK\delta^{-1})}{(1-\gamma)^2}$  for some constant  $C$ ;  $\triangleright$  mini-batch size for inner loop
8:  $\theta^{(0,0)} \leftarrow \{0\} \subset \mathbb{R}^K$   $\triangleright$  initialize the output to contain a single 0-vector


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9: Iterates:
10:  $\triangleright$  Outer loop
11: for  $i = 0, 1, \dots, R'$  do
12:   for each  $k \in [K]$  do
13:     Obtain state samples  $x_k^{(1)}, x_k^{(2)}, \dots, x_k^{(m)} \in \mathcal{S}$  from  $P(\cdot | s_k, a_k)$  for  $(s_k, a_k) \in \mathcal{K}$ . Let
        $w^{(i,0)}(k) \leftarrow \frac{1}{m} \sum_{\ell=1}^m V_{\theta^{(i,0)}}(x_k^{(\ell)}), \quad z^{(i,0)}(k) \leftarrow \frac{1}{m} \sum_{\ell=1}^m V_{\theta^{(i,0)}}^2(x_k^{(\ell)})$   $\triangleright$  empirical estimate of  $P_{\mathcal{K}} V_{\theta^{(i,0)}}$  and  $P_{\mathcal{K}} V_{\theta^{(i,0)}}^2$ 
        $\sigma^{(i,0)}(k) \leftarrow z^{(i,0)}(k) - (w^{(i,0)}(k))^2$   $\triangleright$  empirical estimate of variance  $P_{\mathcal{K}} V_{\theta^{(i,0)}}^2 - (P_{\mathcal{K}} V_{\theta^{(i,0)}})^2$ 
        $\epsilon^{(i,0)}(k) \leftarrow \Theta\left[\sqrt{\log[R'RK\delta^{-1}] \cdot \sigma^{(i,0)}(k) \cdot m^{-1} + \log[R'RK\delta^{-1}](1 - \gamma)^{-1}/m^{3/4}}\right]$   $\triangleright$  estimate of the confidence bound of the empirical estimator  $w^{(i,0)}$ 
        $\bar{w}^{(i,0)}(k) \leftarrow \max\{0, \min\{w^{(i,0)}(k) - \epsilon^{(i,0)}(k), (1 - \gamma)^{-1}\}\}$   $\triangleright$  shift and clip the estimate
14:   end for


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15:    $\triangleright$  Inner loop
16:   for  $j = 1, 2, \dots, R$  do
17:     for each  $k \in [K]$  do
18:       Obtain state samples  $x_k^{(1)}, x_k^{(2)}, \dots, x_k^{(m_1)} \in \mathcal{S}$  from  $P'(\cdot | s_k, a_k)$  for  $(s_k, a_k) \in \mathcal{K}$ . Let
          $w^{(i,j)}(k) \leftarrow \frac{1}{m_1} \sum_{\ell=1}^{m_1} (V_{\theta^{(i,j-1)}}(x_k^{(\ell)}) - V_{\theta^{(i,0)}}(x_k^{(\ell)})) + w^{(i,0)}(k)$   $\triangleright$  empirical estimate of  $P_{\mathcal{K}} V_{\theta^{(i,j-1)}}$ 
          $\epsilon^{(i,j)}(k) \leftarrow \epsilon^{(i,0)}(k) + \Theta(1 - \gamma)^{-1} 2^{-j} \sqrt{\log(RR'K\delta^{-1})/m_1}$   $\triangleright$  approximate the confidence bound of  $P_{\mathcal{K}} V_{\theta^{(i,j-1)}}$ 
          $\bar{w}^{(i,j)}(k) \leftarrow \max\{0, \min\{w^{(i,j)}(k) - \epsilon^{(i,j)}(k), (1 - \gamma)^{-1}\}\}$   $\triangleright$  shift and clip the estimate
19:       end for
20:        $\theta^{(i,j)} \leftarrow \theta^{(i,j-1)} \cup \{\bar{w}^{(i,j)}\}$   $\triangleright$  attach the newly estimated parameter to  $\theta$ 
21:     end for
22:      $\theta^{(i+1,0)} \leftarrow \theta^{(i,R)}$   $\triangleright$  prepare the next outer loop
23:   end for
24: Return  $\theta^{(R',R)}$ 
    
```

Optimal Phased Parametric Q-Learning



Algorithm 2 Optimal Phased Parametric Q-Learning (OPPQ-Learning)

- 1: **Input:** A DMDP $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, r, \gamma)$ with anchor state-action pairs $\mathcal{K}$; feature map $\phi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$;
- 2: **Input:** $\epsilon, \delta \in (0, 1)$
- 3: **Output:** $\theta \subset \mathbb{R}^K$ with $|\theta| = \Theta[(1 - \gamma)^{-1} \log^2 \epsilon^{-1}]$
- 4:
- 5: **Initialize:** $R' \leftarrow \Theta(\log[\epsilon^{-1}(1 - \gamma)^{-1}])$, $R \leftarrow \Theta[R'(1 - \gamma)^{-1}]$ ▷ initialize the numbers of iterations
- 6: $\{w^{(i,j)}, \epsilon^{(i,j)}, \bar{w}^{(i,j)}\}_{i \in [0, R'], j \in [0, R]} \subset \mathbb{R}^K$ as 0 vectors ▷ initialize parameters
- 7: $m \leftarrow C \cdot \frac{1}{\epsilon^2} \cdot \frac{\log(R'RK\delta^{-1})^{4/3}}{(1-\gamma)^3}$, ▷ mini-batch size for outer loop
 $m_1 \leftarrow C \cdot \frac{\log(R'RK\delta^{-1})}{(1-\gamma)^2}$ for some constant C ; ▷ mini-batch size for inner loop
- 8: $\theta^{(0,0)} \leftarrow \{0\} \subset \mathbb{R}^K$ ▷ initialize the output to contain a single 0-vector

Optimal Phased Parametric Q-Learning



9: **Iterates:**

10: $\triangleright$ Outer loop

11: **for** $i = 0, 1, \dots, R'$ **do**

12: **for each** $k \in [K]$ **do**

13: Obtain state samples $x_k^{(1)}, x_k^{(2)}, \dots, x_k^{(m)} \in \mathcal{S}$ from $P(\cdot | s_k, a_k)$ for $(s_k, a_k) \in \mathcal{K}$. Let

$$w^{(i,0)}(k) \leftarrow \frac{1}{m} \sum_{\ell=1}^m V_{\theta^{(i,0)}}(x_k^{(\ell)}), \quad z^{(i,0)}(k) \leftarrow \frac{1}{m} \sum_{\ell=1}^m V_{\theta^{(i,0)}}^2(x_k^{(\ell)})$$

$\triangleright$ empirical estimate of $P_{\mathcal{K}} V_{\theta^{(i,0)}}$ and $P_{\mathcal{K}} V_{\theta^{(i,0)}}^2$

$$\sigma^{(i,0)}(k) \leftarrow z^{(i,0)}(k) - (w^{(i,0)}(k))^2$$

$\triangleright$ empirical estimate of variance $P_{\mathcal{K}} V_{\theta^{(i,0)}}^2 - (P_{\mathcal{K}} V_{\theta^{(i,0)}})^2$

$$\epsilon^{(i,0)}(k) \leftarrow \Theta \left[\sqrt{\log[R'RK\delta^{-1}] \cdot \sigma^{(i,0)}(k) \cdot m^{-1} + \log[R'RK\delta^{-1}](1-\gamma)^{-1}/m^{3/4}} \right]$$

$\triangleright$ estimate of the confidence bound of the empirical estimator $w^{(i,0)}$

$$\bar{w}^{(i,0)}(k) \leftarrow \max \{0, \min \{w^{(i,0)}(k) - \epsilon^{(i,0)}(k), (1-\gamma)^{-1}\}\}$$

$\triangleright$ shift and clip the estimate

14: **end for**

Optimal Phased Parametric Q-Learning



```
15:  ▷ Inner loop
16:  for  $j = 1, 2, \dots, R$  do
17:    for each  $k \in [K]$  do
18:      Obtain state samples  $x_k^{(1)}, x_k^{(2)}, \dots, x_k^{(m_1)} \in \mathcal{S}$  from  $P'(\cdot | s_k, a_k)$  for  $(s_k, a_k) \in \mathcal{K}$ . Let
        
$$w^{(i,j)}(k) \leftarrow \frac{1}{m_1} \sum_{\ell=1}^m \left( V_{\theta^{(i,j-1)}}(x_k^{(\ell)}) - V_{\theta^{(i,0)}}(x_k^{(\ell)}) \right) + w^{(i,0)}(k) \quad \triangleright \text{empirical estimate of } P_{\mathcal{K}} V_{\theta^{(i,j-1)}}$$

        
$$\epsilon^{(i,j)}(k) \leftarrow \epsilon^{(i,0)}(k) + \Theta(1 - \gamma)^{-1} 2^{-i} \sqrt{\log(RR'K\delta^{-1})/m_1}$$

        
$$\bar{w}^{(i,j)}(k) \leftarrow \max \left\{ 0, \min \left\{ w^{(i,j)}(k) - \epsilon^{(i,j)}(k), (1 - \gamma)^{-1} \right\} \right\} \quad \triangleright \text{approximate the confidence bound of } P_{\mathcal{K}} V_{\theta^{(i,j-1)}} \text{ and shift and clip the estimate}$$

19:    end for
20:     $\theta^{(i,j)} \leftarrow \theta^{(i,j-1)} \cup \{\bar{w}^{(i,j)}\}$  ▷ attach the newly estimated parameter to  $\theta$ 
21:  end for
22:   $\theta^{(i+1,0)} \leftarrow \theta^{(i,R)}$  ▷ prepare the next outer loop
23: end for
24: Return  $\theta^{(R',R)}$ 
```

Sample Complexity



Theorem 3 (Near-Optimal Sample Complexity). Suppose $M = (\mathcal{S}, \mathcal{A}, P, r, \gamma)$ is an MDP instance admitting the feature representation $\phi: \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^K$. Suppose that Anchor Assumption holds. Let $\delta, \epsilon \in (0, 1)$ be parameters. The algorithm takes

$$N = \Theta\left[\frac{K}{(1-\gamma)^3 \epsilon^2} \cdot \log^{\frac{4}{3}} \frac{K}{\epsilon \delta (1-\gamma)} \cdot \log^2 \frac{1}{\epsilon (1-\gamma)}\right]$$

samples and outputs θ such that π_θ is ϵ -optimal from every initial state with probability at least $1 - \delta$.

Theorem 1 suggests that, in order to solve the feature-based MDP to precision level ϵ with probability at least $2/3$, any algorithm needs at least $\tilde{\Omega}\left(\frac{K}{(1-\gamma)^3 \epsilon^2}\right)$ sample transitions in the worst case.

Thanks!
